

Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

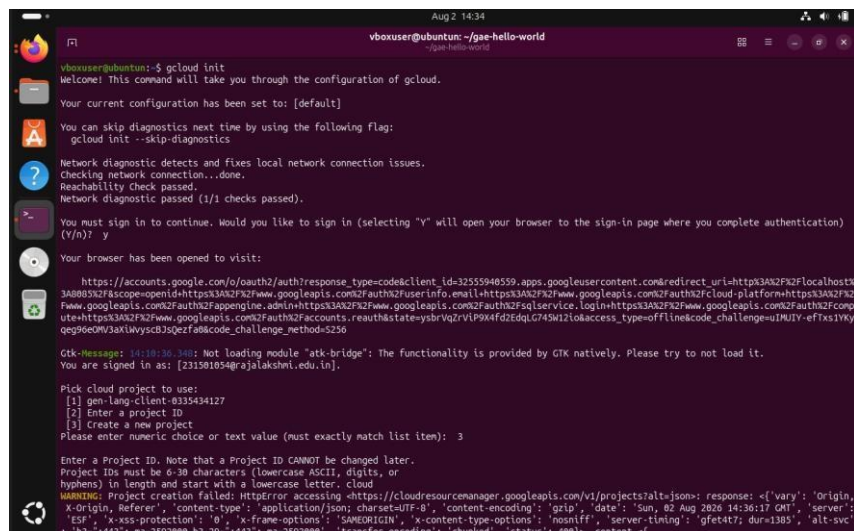
Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-linux-x86_64.tar.gz
tar -xzf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

gcloud init

Sign in to your Google account when prompted.



```
Aug 2 14:34
vboxuser@ubuntu:~/gae-hello-world
vboxuser@ubuntu:~$ gcloud init
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).

You must sign in to continue. Would you like to sign in (selecting "Y" will open your browser to the sign-in page where you complete authentication)
(Y/n)? y

Your browser has been opened to visit:
https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555948559.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%3A8080%2F&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth+state=y5brVqzrVlP9x4fd2EdqL745M12lo&access_type=offline&code_challenge=IMU1Y-eFxs1Vkyqeg96OMV3x1WvysCB35gezfA8&code_challenge_method=S256

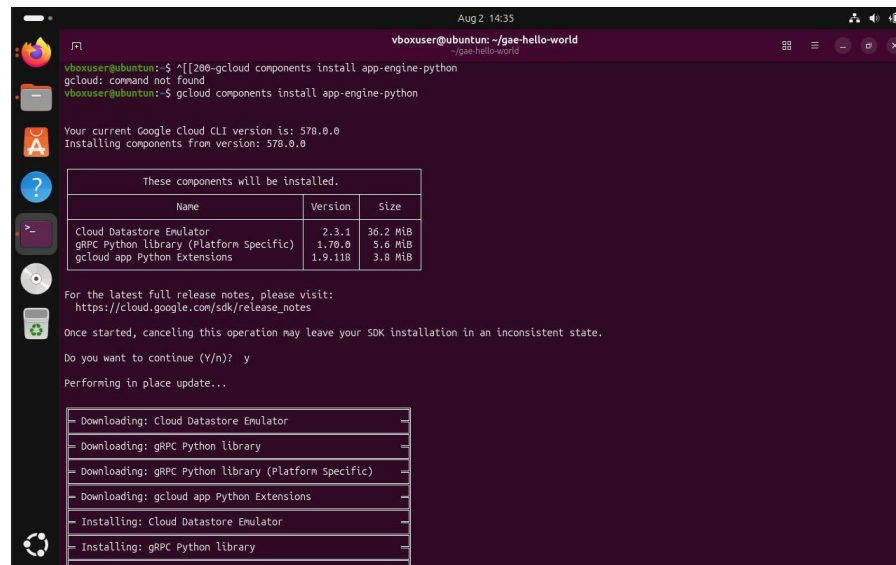
Ctk-Message: 14:36:30: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
You are signed in as: [231501054@rajalakshmi.edu.in].

Pick cloud project to use:
[1] gen-lang-client-8335434127
[2] Enter a project ID
[3] Create a new project
Please enter numeric choice or text value (must exactly match list item): 3

Enter a Project ID. Note that a Project ID CANNOT be changed later.
Project IDs must be 6-30 characters (lowercase ASCII, digits, or
hyphens) in length and start with a lowercase letter, cloud
WARNING: Project creation failed: HttpError accessing https://cloudresourcemanager.googleapis.com/v1/projects?alt=json>: response: <{'vary': 'Origin,
X-Origin, Referer', 'content-type': 'application/json; charset=UTF-8', 'content-encoding': 'gzip', 'date': 'Sun, 02 Aug 2020 14:36:17 GMT', 'server':
'ESF', 'x-oss-protection': '0', 'x-frame-options': 'SAMEORIGIN', 'x-content-type-options': 'nosniff', 'server-timing': 'gfe4t7; dur=1385', 'alt-svc':
'alt=431; ma=3600000 h1=28; c431=ma=3600000', 'transfer-encoding': 'chunked', 'status': 400}>: content of
```

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Monish S

gcloud components install app-engine-python



Step 3: Create the App Directory

```
mkdir gae-hello-world cd gae-  
hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask app =  
Flask(__name__)  
  
@app.route("/") def  
hello():  
    return "Hello World from Google App Engine!"  
  
if __name__ == "__main__": app.run(host="127.0.0.1", port=8080,  
    debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39 handlers:  
- url: /* script: auto
```

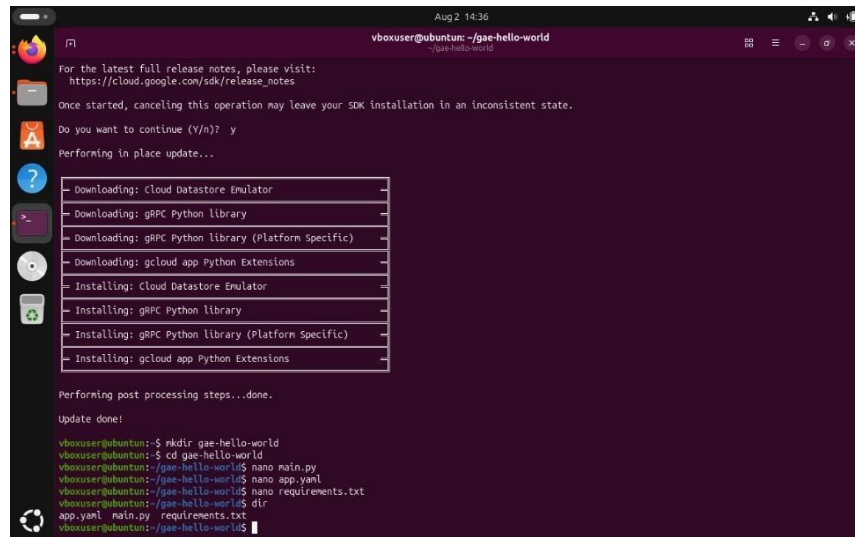
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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The image shows a terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' with a timestamp of 'Aug 2 14:36'. The terminal output includes instructions to visit the release notes, a confirmation to continue, and a progress bar for installing the Google App Engine SDK. The progress bar shows the following steps: Downloading: Cloud Datastore Emulator, Downloading: gRPC Python library, Downloading: gRPC Python library (Platform Specific), Downloading: gcloud app Python Extensions, Installing: Cloud Datastore Emulator, Installing: gRPC Python library, Installing: gRPC Python library (Platform Specific), and Installing: gcloud app Python Extensions. After the installation is complete, the terminal shows the command 'Update done!' and a series of commands to create a directory, change to it, and create files: 'mkdir gae-hello-world', 'cd gae-hello-world', 'nano main.py', 'nano app.yaml', 'nano requirements.txt', 'dir', and 'app.yaml main.py requirements.txt'.

```
vboxuser@ubuntu: ~/gae-hello-world
Aug 2 14:36

For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, canceling this operation may leave your SDK installation in an inconsistent state.

Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu: $ mkdir gae-hello-world
vboxuser@ubuntu: $ cd gae-hello-world
vboxuser@ubuntu: ~/gae-hello-world$ nano main.py
vboxuser@ubuntu: ~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu: ~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$ dir
app.yaml main.py requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.